

Figure 1 (Objective iii) – Normal QQ Plot of Standardised IOP Estimates

$$z = \frac{\hat{\omega}_r - \omega_r}{SE_{JK}} \mid \omega_r = 0.3571 \mid n_s = 300 \text{ clusters/strat} \mid k = 4 \text{ HH/cluster} \mid B = 5000 \text{ reps}$$

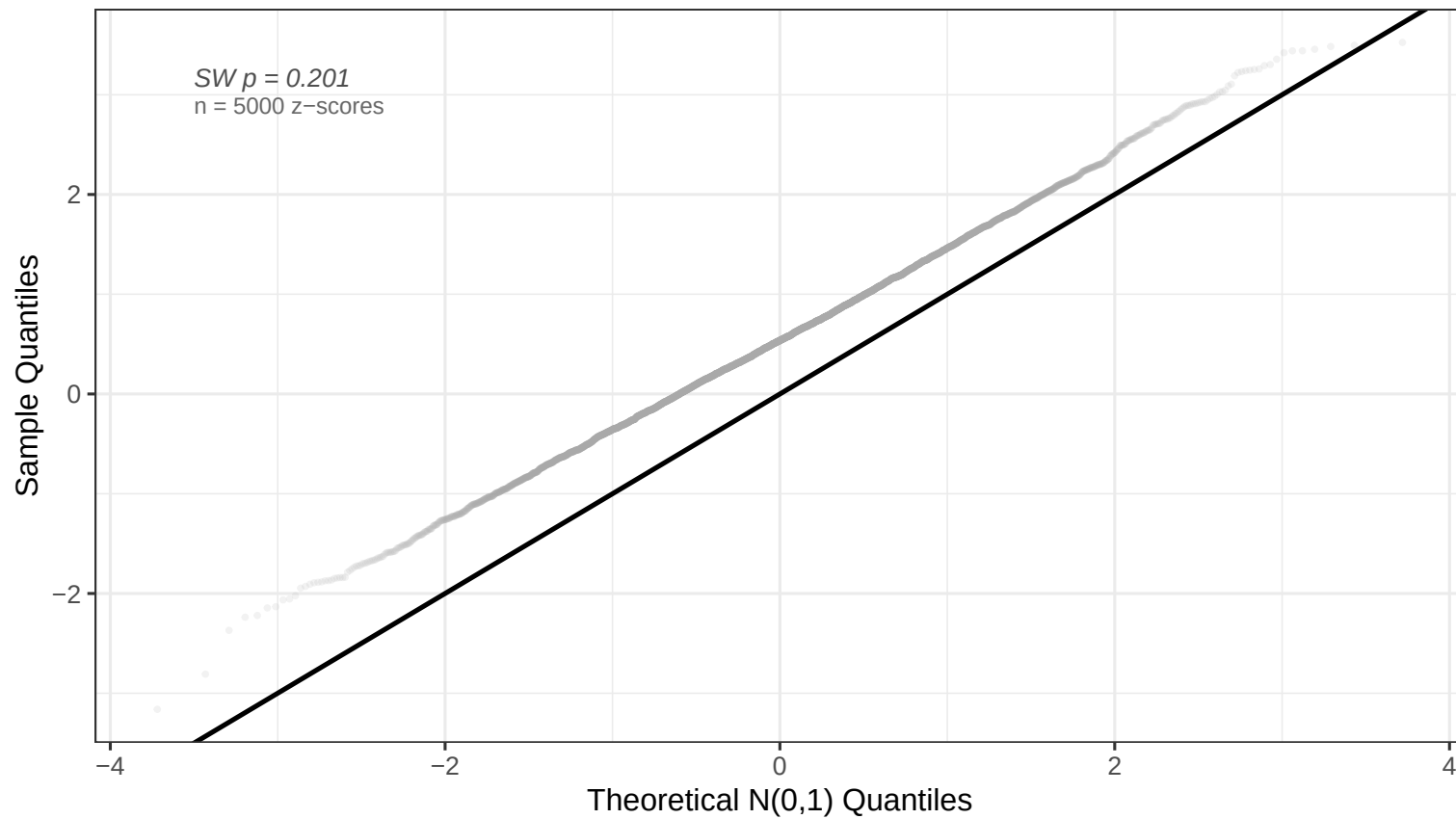


Figure 2 (Objective i) – Sampling Distribution of the Relative IOP Estimator

Bias = 0.01048 | RMSE = 0.02068 | MC SD = 0.01783 | B = 5000 reps

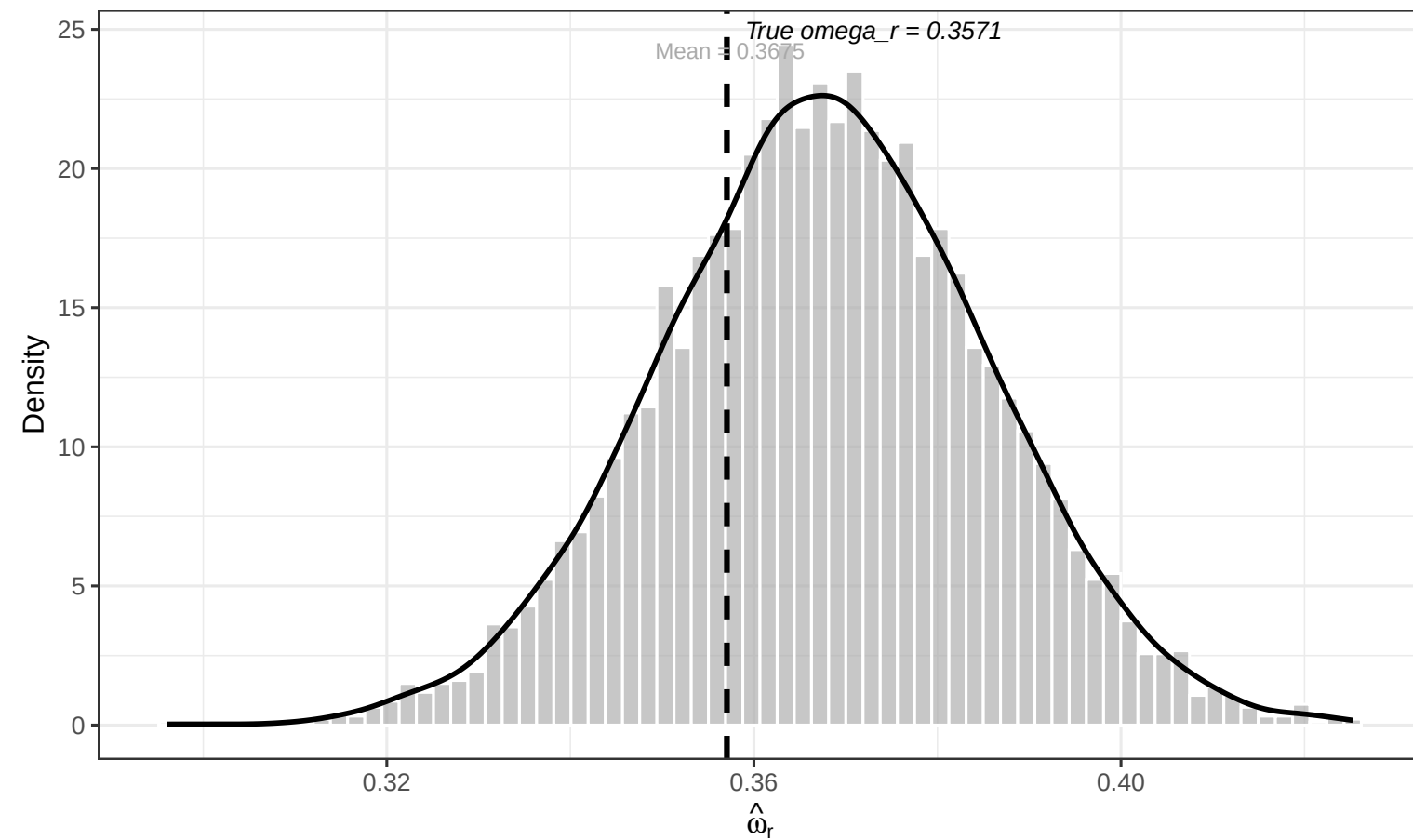


Figure 3 (Objective ii) – Empirical Coverage of 95% Wald Confidence Intervals

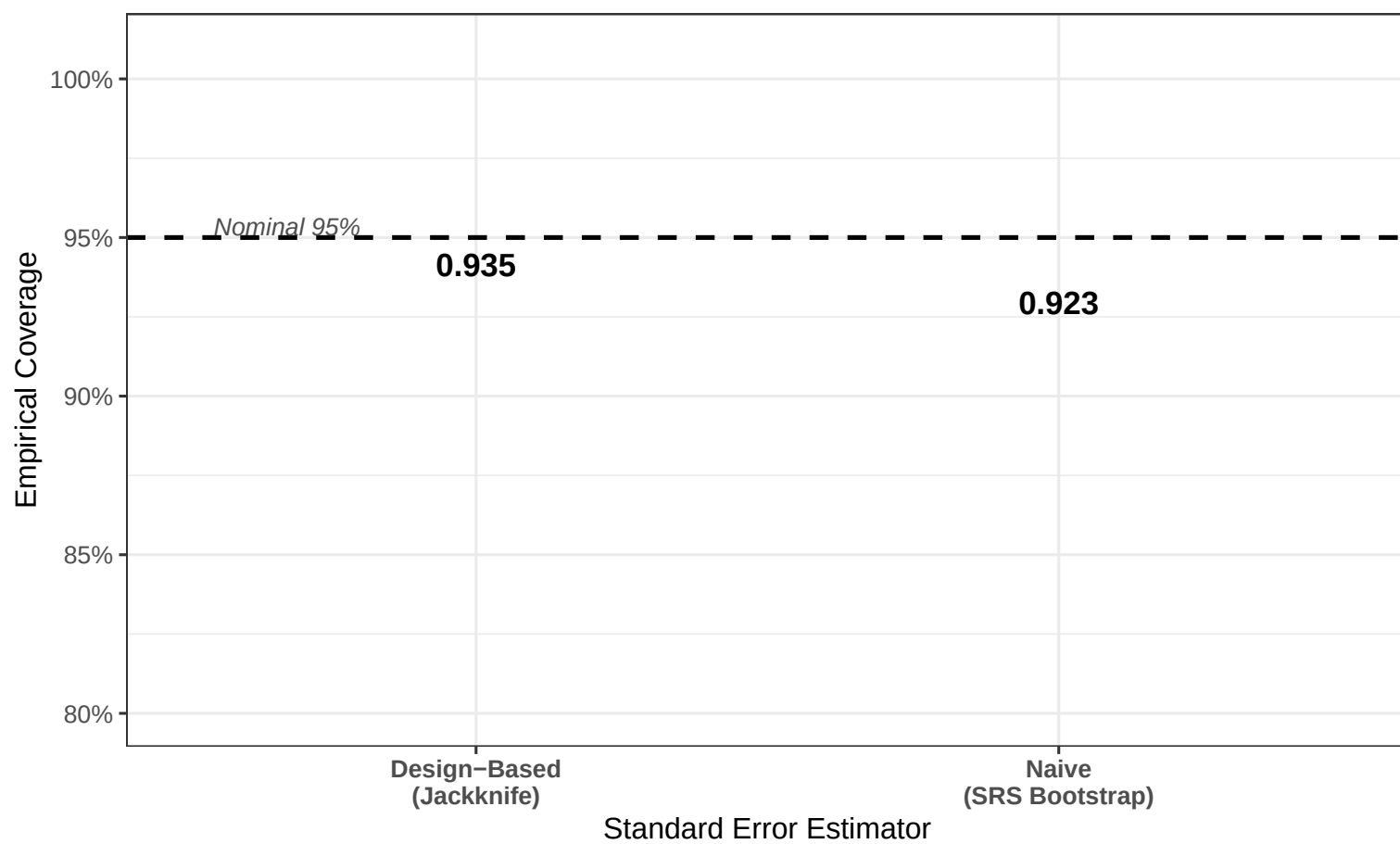
B = 5000 reps | $n_s = 300$ clusters/strat | $k = 4$ HH/cluster

Figure 4 (Objective iv) – Type I Error and Power of the Two-Sample Wald Test

 $H_0 : \omega_r^{(1)} = \omega_r^{(2)}$ vs $H_1 : \omega_r^{(1)} \neq \omega_r^{(2)}$ | $\alpha = 0.05$ | $n_s = 300$ clusters/strat per sample | B = 2000 reps